

IOT-BASED HEALTH MONITORING SYSTEM

*Minor project-1 report submitted
in partial fulfillment of the requirement for award of the degree of*

**Bachelor of Technology
in
Artificial Intelligence and Machine Learning**

By

**G.VISHNU VARDHAN (23UEAM0020) (VTU 25007)
B.LOKESH (23UEAM0006) (VTU 25350)**

*Under the guidance of
Dr. S. ALEX DAVID, B.E, M.E, Ph.D.,
PROFESSOR*



**DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
SCHOOL OF COMPUTING**

**VEL TECH RANGARAJAN DR. SAGUNTHALA R&D INSTITUTE OF
SCIENCE & TECHNOLOGY**

(Deemed to be University Estd u/s 3 of UGC Act, 1956)

**Accredited by NAAC with A++ Grade
CHENNAI 600 062, TAMILNADU, INDIA**

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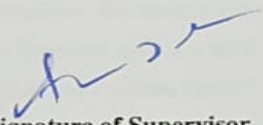
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CERTIFICATE

It is certified that the work contained in the project report titled "IOT-BASED HEALTH MONITORING SYSTEM" by "G.VISHNU VARDHAN (23UEAM0020), B.LOKESH (23UEAM0006), has been carried out under my supervision and that this work has not been submitted elsewhere for a degree.


Signature of Supervisor

Dr. S. Alex David

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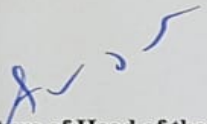
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November, 2025


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Institute of Science & Technology

November , 2025


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Vel Tech Rangarajan Dr. Sagunthala R&D

Institute of Science & Technology

November, 2025



DECLARATION

We declare that this written submission represents my ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.



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G.VISHNU VARDHAN

Date: 13 / 11 / 25



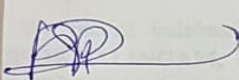
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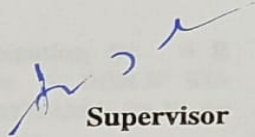
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Date: 13 / 11 / 25

APPROVAL SHEET

This project report entitled "IOT-BASED HEALTH MONITORING SYSTEM" by G.VISHNU VARDHAN (23UEAM0020), B.LOKESH (23UEAM0006), is approved for the degree of B.Tech in Computer Science & Engineering.


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Place:

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ABSTRACT

The Internet of Things (IoT) Based Health Monitoring System is an innovative approach designed to continuously monitor vital health parameters of patients in real time. In this system, various biomedical sensors such as temperature sensors, heart rate sensors, and SpO sensors are used to measure body temperature, pulse rate, and oxygen levels, respectively. The collected data is transmitted to a microcontroller (such as Arduino or ESP32), which processes the signals and sends them to the cloud or a web/mobile application through the Internet using Wi-Fi or GSM modules. This system enables doctors and caregivers to remotely monitor patients' health status from anywhere, thereby reducing the need for frequent hospital visits and enabling timely medical intervention in case of abnormalities. Additionally, alert notifications (via SMS, email, or app) can be generated automatically if any parameter exceeds the predefined threshold values.

The proposed IoT-based health monitoring system provides a cost-effective, efficient, and real-time health supervision solution, especially useful for elderly people, chronic disease patients, and rural healthcare systems. It demonstrates how IoT technology can significantly improve the quality and accessibility of healthcare services. It connects wearable sensors to measure vital parameters such as heart rate, blood pressure, temperature, and oxygen level, which are then transmitted to a cloud server through the internet. Doctors and caregivers can access this data remotely and receive instant alerts if any abnormal readings are detected. This system helps in early diagnosis, continuous monitoring, and timely medical assistance, especially for elderly or chronically ill patients. It reduces hospital visits, saves time and costs, and improves healthcare efficiency by enabling remote and smart health management.

Keywords:

Biomedical Sensors, Cloud Computing, Data Analytics, Health Monitoring System, Internet Of Things, Real-time Monitoring, Remote Patient Monitoring, Smart Healthcare, Wearable Devices, Wireless Sensor Networks (WSN)

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LIST OF ACRONYMS AND ABBREVIATIONS

AI	Artificial Intelligence
BLE	Bluetooth Low Energy
CPU	Central Processing Unit
DHT	Digital Humidity and Temperature (sensor)
ECG	Electrocardiogram
FIFO	First In, First Out
HTTP	Hypertext Transfer Protocol
IOT	Internet of Things
JSON	JavaScript Object Notation
KNN	K-Nearest Neighbor
MQTT	Message Queuing Telemetry Transport
NFC	Near Field Communication
SPO	Peripheral Capillary Oxygen Saturation
TCP/IP	Transmission Control Protocol

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Chapter 1

INTRODUCTION

1.1 Introduction

The IoT-based Health Monitoring System is a modern technological innovation designed to revolutionize the healthcare industry by enabling real-time monitoring, analysis, and management of patients' health conditions through interconnected smart devices. This system utilizes various sensors—such as heart rate, temperature, blood pressure, oxygen saturation, and ECG sensors—connected to an Internet of Things (IoT) network that continuously collects and transmits patient data to cloud based platforms or healthcare providers for remote access. Through IoT integration, doctors and caregivers can monitor patients' vital signs anytime and anywhere, ensuring timely medical intervention in case of abnormalities or emergencies. Additionally, the system minimizes the need for frequent hospital visits, improves patient outcomes, and supports data-driven decision-making by maintaining accurate health records over time. The integration of IoT with artificial intelligence (AI) and big data analytics further enhances the system's capabilities, allowing for predictive diagnosis, personalized healthcare, and automated alerts. Overall, an IoT-based health monitoring system represents a significant advancement toward smart healthcare infrastructure, promoting efficiency, accessibility, and better quality of life for patients across the globe

1.2 Aim of the project

The aim of the IoT-based Health Monitoring System project is to design and develop a smart, efficient, and reliable system that enables continuous and real-time monitoring of patients' vital health parameters using Internet of Things (IoT) technology. The main objective is to remotely collect, analyze, and transmit health data from various sensors—such as temperature, heart rate, blood pressure, and oxygen level sensors—to healthcare professionals or caregivers through a secure network. This system aims to provide early detection of abnormal health conditions, reduce the need for frequent hospital visits, and ensure timely medical intervention in emer

gencies. Additionally, the project seeks to enhance patient care by integrating cloud storage and mobile applications for easy access to medical data, thereby improving the overall efficiency and responsiveness of healthcare services

1.3 Project Domain

The project domain for the IoT-based Health Monitoring System falls under the Internet of Things (IoT) and Healthcare Technology domains. It combines principles of embedded systems, wireless sensor networks, cloud computing, and data analytics to create a smart healthcare solution. This domain focuses on developing connected medical devices and applications that can monitor, collect, and analyze patients' health data in real time. It also overlaps with areas such as Biomedical Engineering, Telemedicine, and Artificial Intelligence in Healthcare, where IoT technologies are used to improve patient care, enhance diagnostic accuracy, and enable remote health management. Overall, the project lies at the intersection of IoT and Healthcare Informatics, aiming to make medical services more efficient, accessible, and proactive through smart automation and connectivity

1.4 Scope of the Project

The scope of the IoT-based Health Monitoring System project is extensive and covers various aspects of modern healthcare and technology integration. This project aims to develop a smart, real-time health tracking system that continuously monitors patients' vital parameters—such as heart rate, body temperature, blood pressure, and oxygen saturation—using IoT-enabled sensors. The collected data is transmitted to a cloud server or mobile application, where healthcare professionals and family members can access it remotely. The system can generate automatic alerts or notifications in case of any abnormal readings, ensuring timely medical response and reducing the risk of severe health complications. The project's scope extends to both clinical and home healthcare environments, making it useful for hospitals, elderly care, remote patients, and individuals with chronic conditions. It also supports scalability, allowing integration with artificial intelligence (AI) for predictive analysis and machine learning models for disease detection. Additionally, the system can be expanded to include wearable technology, GPS tracking for patient mobility, and blockchain based data security for protecting sensitive health information. Overall, the scope of

this project highlights its potential to improve healthcare accessibility, reduce costs, enhance patient safety, and pave the way for smarter, technology-driven medical systems in the future

Chapter 2

LITERATURE REVIEW

2.1 Literature Review

Integration of the Internet of Things (IoT) with healthcare technologies has gained significant attention due to its potential to revolutionize patient monitoring and management. According to Patel et al. (2020), IoT-based health monitoring systems provide continuous observation of vital signs and enable remote healthcare services, thereby reducing hospital visits and medical costs. Kumar and Raj (2019) developed an IoT-enabled system that uses wearable sensors to track heart rate and temperature, transmitting real-time data to cloud servers for medical analysis. Similarly, Babu et al. (2021) proposed a system utilizing the ESP8266 microcontroller and ThingSpeak platform to monitor patient vitals and generate alerts when abnormal readings are detected. Studies such as Sharma et al. (2020) highlight the importance of wireless communication technologies like Wi-Fi and Bluetooth in ensuring efficient data transmission between sensors and healthcare providers. Moreover, advancements in cloud computing and mobile applications have enhanced accessibility, enabling doctors to monitor patients remotely and respond promptly during emergencies. Overall, the reviewed literature emphasizes that IoT-based health monitoring systems significantly improve healthcare quality, patient safety, and disease management through real-time data collection, analysis, and communication.

2.2 Gap Identification

Although numerous studies have explored IoT-based health monitoring systems, several gaps remain that limit their widespread implementation and effectiveness. Most existing systems focus on monitoring only a few health parameters, such as temperature or heart rate, without integrating multiple vital signs into a single platform for comprehensive analysis. In addition, many prototypes rely on cloud platforms for data storage but lack robust security mechanisms, making patient data vulnerable to privacy breaches. Another significant gap is the limited use of real-time alert systems that can promptly notify doctors or family members during emergen-

cies. Furthermore, existing systems often require continuous internet connectivity, which restricts their use in rural or low-connectivity areas. Many studies also fail to incorporate data analytics or artificial intelligence for predictive health assessment, which could enhance decision-making and early diagnosis. Therefore, there is a need for a low-cost, secure, and reliable IoT-based health monitoring system capable of real-time data collection, analysis, and alert generation, even in low-resource environments.

Chapter 3

PROJECT DESCRIPTION

3.1 Existing System

An existing IoT-based health monitoring system is the ESP32 and ThingSpeak-based Patient Health Monitoring System, widely used in academic and prototype healthcare projects. This system continuously monitors a patient's vital signs such as heart rate, body temperature, and oxygen saturation (SpO₂) using sensors like the MAX30100 pulse oximeter and LM35 temperature sensor. The sensors are connected to an ESP32 microcontroller, which has built-in Wi-Fi capability to transmit data to the ThingSpeak cloud platform. Existing health monitoring systems, whether traditional or partially IoT-enabled, face several limitations that reduce their effectiveness and reliability. Most conventional systems rely on manual data collection or hospital-based monitoring, which restricts continuous observation of patients' vital signs. As a result, any sudden changes in health conditions often go unnoticed until the next scheduled check-up, leading to delayed medical responses. Many existing IoT-based health monitoring systems also suffer from limited accuracy and reliability due to the use of low-quality sensors or poor calibration. In addition, some systems lack real-time data transmission capabilities or stable internet connectivity, especially in rural or low-network areas, making remote monitoring ineffective. Data privacy and security are also major concerns, as sensitive health information may be vulnerable to unauthorized access.

3.2 Problem statement

Demand for efficient and continuous health monitoring has increased due to the rise in chronic diseases, aging populations, and the need for real-time medical attention. Traditional healthcare systems primarily depend on manual monitoring and periodic check-ups, which often fail to detect sudden changes in patients' health conditions. This delay in diagnosis and response can lead to serious health compli-

cations or even fatalities. Furthermore, in rural and remote areas, access to quality healthcare services is limited, making it difficult for patients to receive timely medical attention. To address these challenges, there is a need for an IoT-based health monitoring system capable of continuously measuring vital parameters such as heart rate, body temperature, blood pressure, and oxygen levels. In recent years, the demand for efficient and continuous health monitoring has increased due to the rise in chronic diseases, aging populations, and the need for real time medical attention. Traditional healthcare systems primarily depend on manual monitoring and periodic check-ups, which often fail to detect sudden changes in patients' health conditions. This delay in diagnosis and response can lead to serious health complications or even fatalities. Furthermore, in rural and remote areas, access to quality healthcare services is limited, making it difficult for patients to receive timely medical attention. To address these challenges, there is a need for an IoT-based health monitoring system capable of continuously measuring vital parameters such as heart rate, body temperature, blood pressure, and oxygen levels.

3.3 System Specification

3.3.1 Hardware Specification

1. Microcontroller:

- ESP8266 / ESP32 (for Wi-Fi connectivity and IoT data transmission)

2. Sensors:

- Heart Rate Sensor, e.g., Pulse Sensor or MAX30100/MAX30102 Measure-pulserate
- oxygen saturation (SpO)
- Temperature Sensor (e.g., LM35 or DS18B20)– Monitors body temperature
- Blood Pressure Sensor ,e.g., BMP180 or similar cuff-based sensor, Measures
- systolic and diastolic pressure
- •OxygenSaturation Sensor,SpOsensor, e.g., MAX30102 Measuresblood-oxygen

3. Power Supply:

- 5V DC adapter or rechargeable Li-ion battery

4. Display Module (optional):

- LCD or OLED display for local data visualization

5. Connectivity Modules:

- Wi-Fi (ESP8266/ESP32 integrated) for cloud communication

6. Optional:

- Bluetooth module (HC-05/HC-06) for local device communication

7. Miscellaneous Components:

- Jumper wires, resistors, breadboard or PCB
- Enclosure for compact hardware assembly

3.3.2 Software Specification

1. Programming Languages:

- C/C++ (for IoT).
- Python/Node.js (backend).
- JavaScript/Dart (frontend).

2. Frameworks:

- Flask / Django / Express.js / React.

3. Cloud Platforms:

- AWS / Firebase / ThingSpeak.

5. Databases:

- MySQL / MongoDB / Firebase Realtime DB.

6. Version Control:

- Git, GitHub.

7. Development Environment:

- Arduino IDE, VS Code

3.3.3 Standards and Policies

Sample attached

Anaconda Prompt

Anaconda prompt is a type of command line interface which explicitly deals with the ML(MachineLearning) modules.And navigator is available in all the Windows,Linux and MacOS.The anaconda prompt has many number of IDE's which make the coding easier. The UI can also be implemented in python.

Standard Used: ISO/IEC 27001

Jupyter

It's like an open source web application that allows us to share and create the documents which contains the live code, equations, visualizations and narrative text. It can be used for data cleaning and transformation, numerical simulation, statistical modeling, data visualization, machine learning.

Standard Used: ISO/IEC 27001

Chapter 4

METHODOLOGY

4.1 Proposed System

Proposed IoT-based Health Monitoring System is designed to provide a reliable, real-time, and continuous monitoring solution for patients' vital health parameters using the integration of advanced sensors, microcontrollers, cloud computing, and IoT communication technologies. The system primarily aims to overcome the limitations of conventional healthcare systems by automating the process of health data acquisition, analysis, and remote transmission, thereby enhancing the efficiency and accessibility of medical care. In this proposed model, multiple physiological parameters such as heart rate, body temperature, oxygen saturation (SpO₂), blood pressure, and electrocardiogram (ECG) readings are measured using wearable and non-invasive biomedical sensors connected to an IoT-enabled microcontroller like Arduino Uno, ESP32, or Raspberry Pi. These sensors collect real-time data from the patient's body and transmit it wirelessly through Wi-Fi, Bluetooth, or GSM modules to a cloud-based platform for storage and processing. Once the data is uploaded to the cloud, it can be visualized and analyzed using web based or mobile dashboard applications developed using platforms such as ThingSpeak, Blynk, or Firebase. The system ensures continuous monitoring by periodically updating the sensor data, allowing healthcare professionals and family members to track the patient's health remotely at any time. The use of IoT technology enables real-time alerts through SMS, email, or push notifications whenever any abnormal health condition is detected—for example, a sudden drop in oxygen level, a rise in temperature, or irregular heartbeat patterns. This alert mechanism facilitates quick medical intervention and can potentially save lives, especially in cases of chronic illness or emergency conditions.

4.2 General Architecture



Figure 4.1: General Architecture

Diagram illustrates a comprehensive cloud-based healthcare monitoring system that integrates medical sensor networks, cloud computing, and end-user applications to provide efficient, real-time, and intelligent healthcare services. At the core of this architecture lies the Medical Sensor Network, which consists of a set of smart medical sensors attached to different parts of a patient's body. These sensors are responsible for continuously collecting vital physiological signals such as heart rate, body temperature, oxygen saturation, blood pressure, glucose levels, and electrocardiogram (ECG) readings. The collected data represents the patient's health status in real time and provides the foundation for further medical analysis. The sensors operate wirelessly, often using low-energy communication protocols such as Bluetooth, ZigBee, or Wi-Fi to transmit data securely to nearby Gateway Devices. The Gateway Devices act as intermediaries between the body sensors and the cloud infrastructure. These devices, which can be smartphones, tablets, or laptops, gather raw medical data from the sensors, perform initial filtering or pre processing, and then send the information to the cloud through the Internet. The gateway also plays a crucial role in ensuring data security, encryption, and proper communication management before the information is transmitted to the remote cloud servers. Once the sensor data reaches the cloud, it is handled by the Cloud Service Provider, which offers multiple functional layers such as storage, processing, analysis, and verification

4.3 Design Phase

4.3.1 Data Flow Diagram

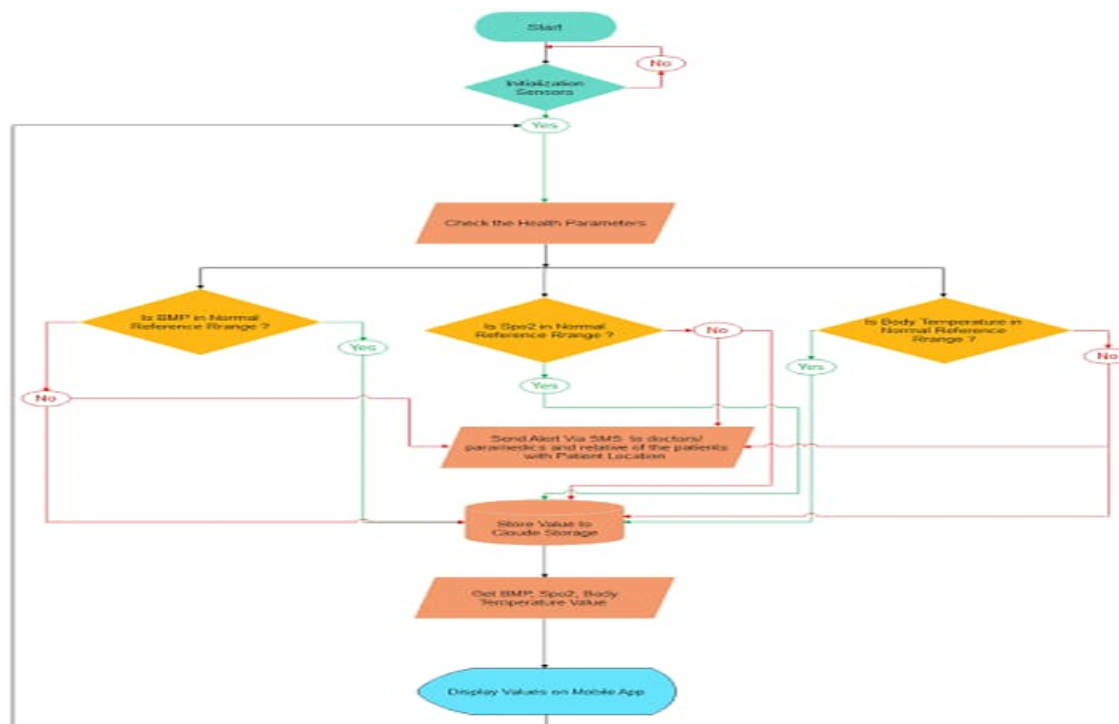


Figure 4.2: Data Flow Diagram

Flowchart illustrates the operational flow of a smart health monitoring system that utilizes body sensors, cloud storage, and mobile applications for real-time patient health tracking. The process begins at the Start node, which marks the initiation of the system. The first step involves the Initialization of Sensors, a critical phase in which the wearable medical sensors attached to the patient's body are powered on and calibrated. These sensors are responsible for collecting vital physiological parameters, including BMP (beats per minute) representing heart rate, SpO (oxygen saturation) levels, and body temperature. The system performs a self-check during initialization to verify that all sensors are functioning correctly and are connected properly. If the sensors fail to initialize, the process loops back, ensuring the system does not proceed with faulty equipment. Once the sensors are successfully initialized, the system transitions to the next stage, where it begins to check and record the health parameters continuously. In this stage, the system retrieves the patient's health readings from the sensors and compares them to standard medical reference ranges. The flowchart then divides into

three main decision branches, each corresponding to one key health parameter: BMP, SpO₂, and body temperature. For each parameter, the system checks whether the recorded value falls within the normal reference range predefined by medical standards.

4.3.2 Use Case Diagram

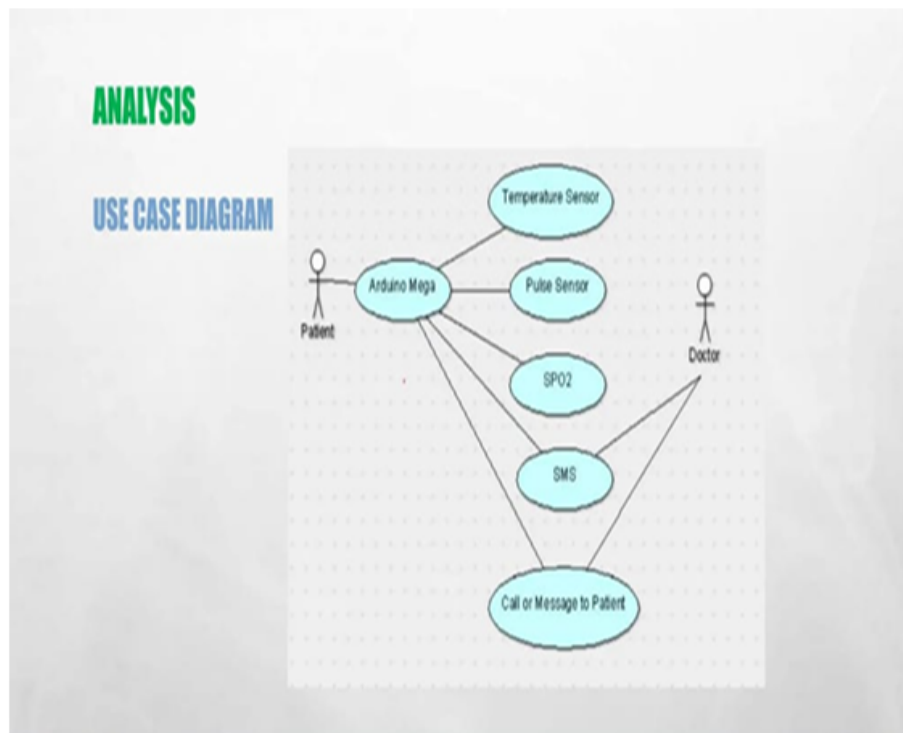


Figure 4.3: Use Case Diagram

Presented diagram illustrates the Use Case Model for a Smart Health Monitoring System that utilizes Arduino Mega as the central controller to monitor patient health parameters and communicate with healthcare professionals. The diagram captures the functional interactions between the actors—namely the Patient and the Doctor—and the system components that enable seamless health data acquisition, analysis, and communication. In this use case, the Patient interacts directly with the Arduino-based health monitoring system, which serves as the central hub for gathering and processing medical data from various sensors. The Arduino Mega, a microcontroller board equipped with multiple input and output pins, plays a crucial role in collecting physiological data, analyzing it, and transmitting relevant information to both the patient and the doctor. At the start of the process, the Patient wears or connects the necessary medical sensors that measure vital signs. The first

use case connected to the Arduino Mega is the Temperature Sensor, which continuously monitors the patient’s body temperature. This sensor is designed to detect fluctuations that might indicate fever, hypothermia, or other health anomalies

4.3.3 Class Diagram

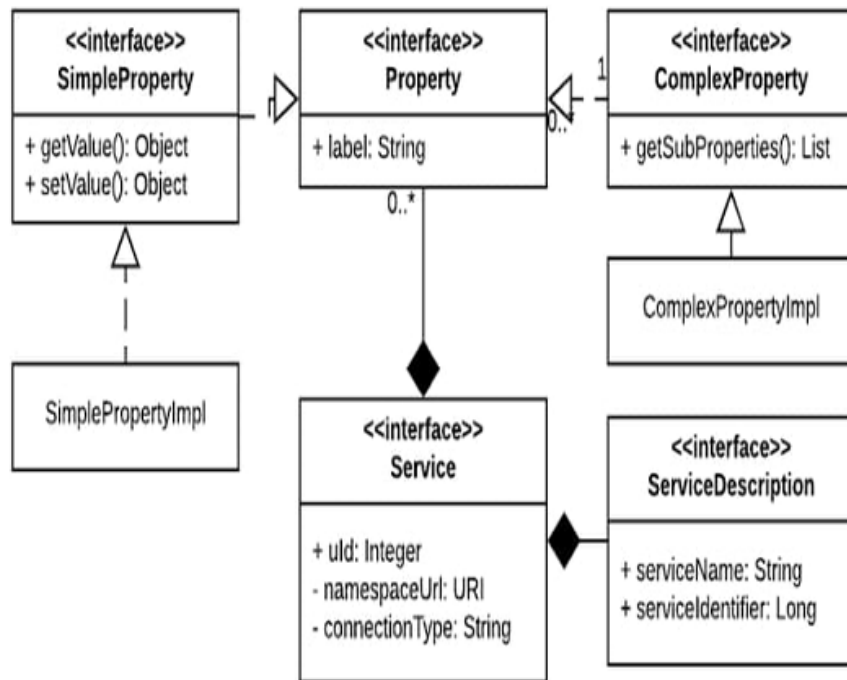


Figure 4.4: Class Diagram

Class diagram representing the conceptual and structural design of an IoT-based Health Monitoring System’s service-property model. It defines how properties and services are represented, implemented, and interconnected in an object-oriented IoT architecture. This model provides a flexible and scalable way to describe the characteristics, behaviors, and interrelationships of devices and their services within an IoT framework —essential for healthcare systems where different medical sensors and devices interact dynamically. At the center of this diagram is the **Property** interface, which forms the foundational abstraction for defining any measurable or configurable aspect of an IoT device. A “Property” represents a feature, characteristic, or measurable value associated with an IoT device, such as temperature, blood pressure, oxygen level, heart rate, or glucose level. It contains one attribute, `label: String`, which identifies or names the property. This ensures every property in the IoT net

work has a clear, human-readable label that can be referenced programmatically or displayed in the user interface

4.3.4 Sequence Diagram

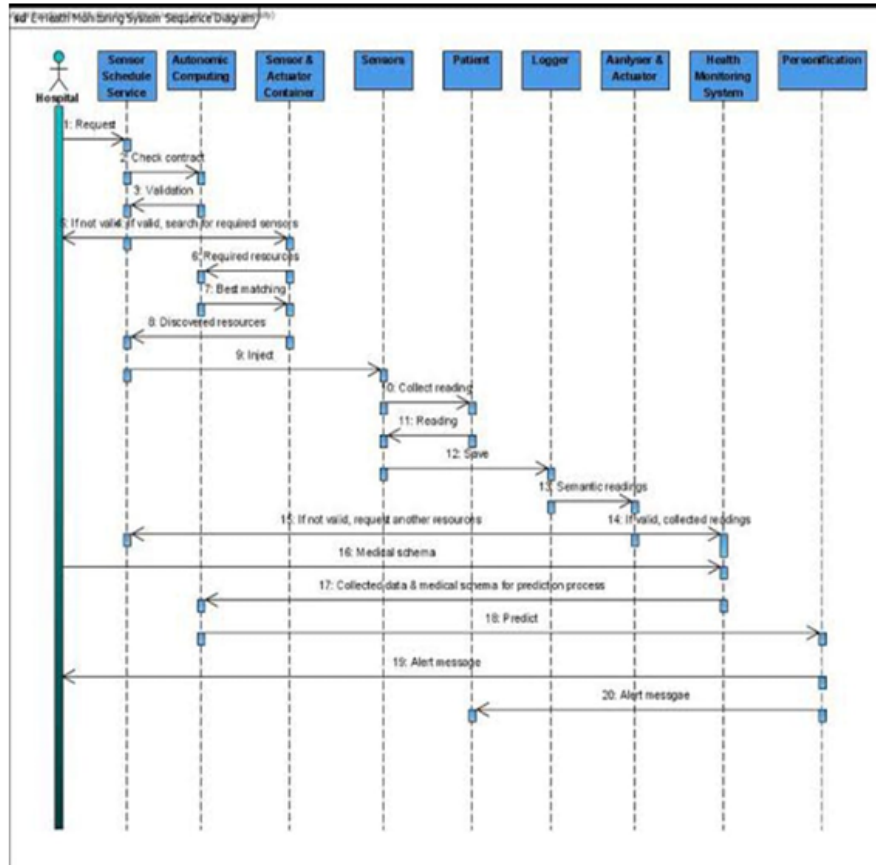


Figure 4.5: Sequence Diagram

IoT-based Health Monitoring System is an intelligent health care framework designed to continuously track, analyze, and manage patient health data using interconnected sensors, cloud services, and analytical algorithms. The process begins when a hospital initiates a request to the system through the Sensor Schedule Service for monitoring a specific patient. Upon receiving this request, the service first checks the contract and validates it to ensure that the hospital has authorized access and that all necessary permissions are in place. If the validation process is successful, the system proceeds to search for the required sensors essential for patient data collection. These sensors may include heart rate monitors, temperature sensors, blood pressure sensors, oxygen saturation detectors, glucose monitors, and other biomedical instruments depending on the patient's condition. Once the relevant

sensors are identified, the Autonomic Computing component becomes active. This module handles intelligent decision-making, resource allocation, and dynamic adaptation. It retrieves the required resources, performs a best-matching operation, and identifies the most suitable sensor configurations through a discovery mechanism. After the discovery process, the Sensor and Actuator Container injects these identified sensors into the monitoring environment, establishing a connection between the patient and the system's analytical backbone

4.3.5 Collaboration diagram

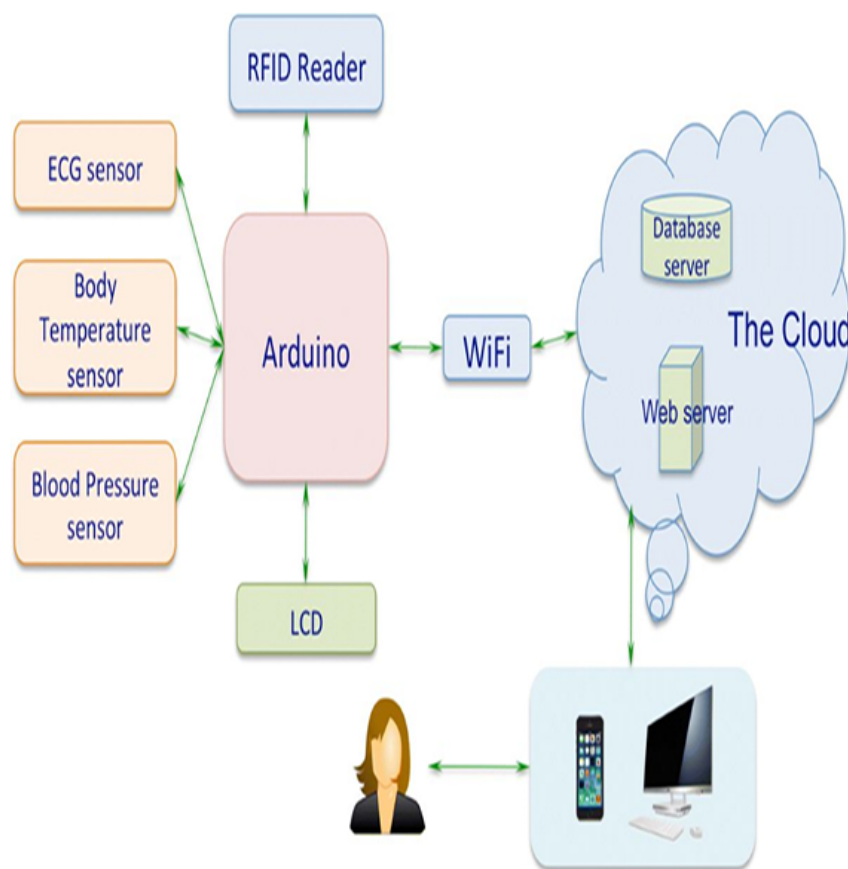


Figure 4.6: Collaboration diagram

Internet of Things (IoT)-based Health Monitoring System is an innovative and intelligent healthcare solution designed to continuously monitor, record, and analyze a patient's vital health parameters in real time. The system integrates various biomedical sensors with embedded hardware, wireless communication modules, and cloud computing infrastructure to create a seamless and automated patient monitoring environment. The central component of this system is the

Arduino microcontroller, which serves as the primary processing unit responsible for collecting, processing, and transmitting data from multiple sensors connected to the patient. The main sensors integrated into this architecture include the Electro-cardio gram (ECG) sensor, the Body Temperature sensor, and the Blood Pressure sensor. Each of these sensors plays a crucial role in detecting vital signs—where the ECG sensor monitors the electrical activity of the heart to detect arrhythmias or cardiac abnormalities, the temperature sensor measures body temperature to detect fever or infection, and the blood pressure sensor records systolic and diastolic pressure to assess circulatory health. The sensors are attached to the patient's body and continuously send analog signals representing physiological conditions to the Arduino board, which converts these signals into digital form through its built-in Analog to-Digital Converter (ADC). The Arduino then processes this data locally and temporarily displays real-time results on an LCD screen, allowing both the patient and nearby medical staff to observe immediate readings

4.3.6 Activity Diagram

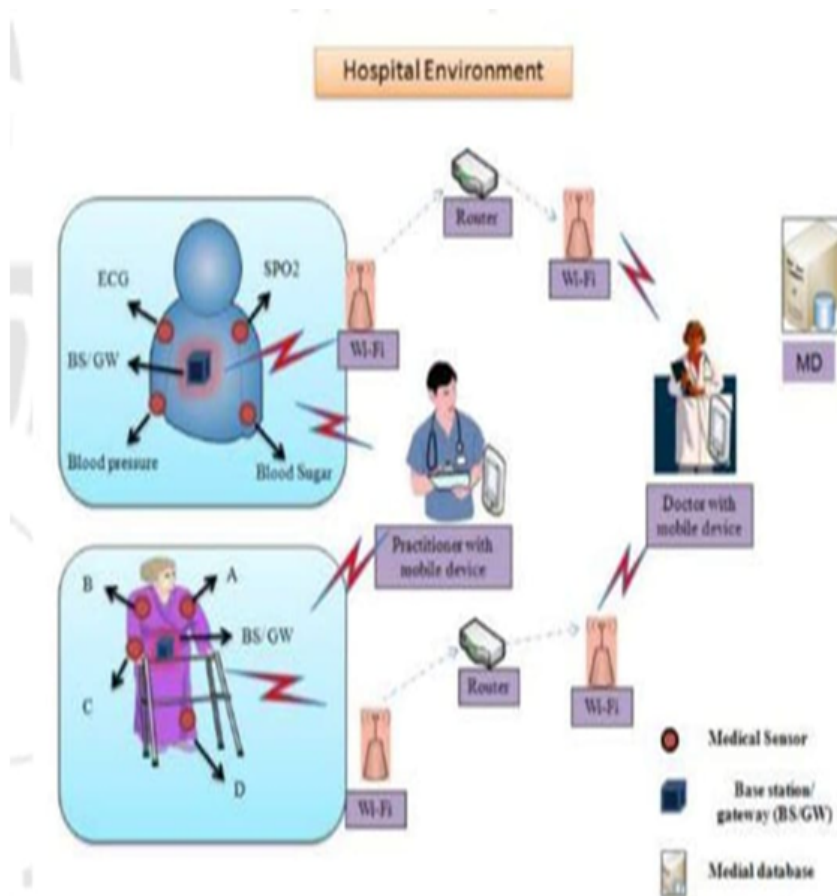


Figure 4.7: Activity Diagram

4.4 Algorithm & Pseudo Code

4.4.1 Algorithm

1. Start
2. Initialize all sensors (e.g., heart rate, temperature, SpO₂, ECG).
3. Establish connection between the IoT device (e.g., Arduino/ESP32) and the cloud server via Wi-Fi or Bluetooth.
4. Continuously read sensor data from the connected sensors.
5. Preprocess the data (filter noise, calculate averages, convert signals to readable values).
6. Display sensor readings on an LCD or mobile application interface.
7. Check threshold conditions for each health parameter:
8. If heart rate, SpO₂, or temperature values go beyond normal range → mark as abnormal.
9. If abnormal data detected:
10. Generate an alert message (e.g., “High Heart Rate” or “Low Oxygen Level”).
11. Send alert to the doctor or caregiver via the cloud or SMS/email.
12. Transmit data periodically to the IoT cloud platform (e.g., Thingspeak, Blynk, Firebase) for storage and monitoring.
13. Visualize health data on a dashboard or mobile app in real time (graphs, charts, etc.).
14. Log the data in the cloud database for further analysis and record keeping.
15. Repeat steps 4–11 continuously while the system is running.
16. If device powered off or stopped, end data collection.
17. Stop

4.4.2 Pseudo Code

```
1
2 BEGIN
3 // Step 1: Initialize system components
4 I n i t i a l i z e HeartRate Sensor
5 I n i t i a l i z e Temperature Sensor
6 I n i t i a l i z e SpO2 Sensor
7 I n i t i a l i z e WiFi Module
8 17
9 I n i t i a l i z e Display Module
10 // Step 2: Connect to cloud or IoT platform
11 Connect WiFi Module to Network
12 Connect Device to Cloud Server
13 // Step 3: Start continuous monitoring loop
14 WHILE (System is ON) DO
15 // Step 4: Read sensor values
16 heart rate
17 Read( HeartRate Sensor )
18 t emperature
19 spo2
20 Read( Temperature Sensor )
21 Read( SpO2 Sensor)
22 // Step 5: Display readings locally
23 Display ( Heart Rate : , heart rate )
24 Display ( Temperature : , temperature )
25 Display ( SpO2 Level : , spo2)
26 // Step 6: Check health condition thresholds
27 IF ( heart rate < 60 OR heart rate > 100) THEN
28 alert message
29 Abnormal Heart Rate Detected !
30 Send Alert ( alert message )
31 ENDIF
32 IF (temperature > 38.0) THEN
33 alert message
34 High Body Temperature Detected !
35 Send Alert ( alert message )
36 ENDIF
37 IF (spo2 < 90) THEN
38 alert message
39 Low Oxygen Level Detected !
40 Send Alert ( alert message )
41 ENDIF
42 // Step 7: Send sensor data to cloud
43 Send To Cloud( heart rate , temperature , spo2)
44 // Step 8: Store or visualize data on cloud dashboard
45 Update Cloud Dashboard ()
46 // Step 9: Wait for next reading cycle
47 Delay (5000)
48 // delay for 5 seconds (example)
```

```

49 END WHILE
50 / / Step 10: End program when system stops
51 Disconnect WiFi Module
52 Stop all sensors
53 END

```

4.4.3 Data Set / Generation of Data (Description only)

Explanation of Fields:

- Patient ID: Unique identifier for each individual.
- Timestamp: Time of data collection by IoT sensors.
- Heart Rate (BPM): Beats per minute measured by pulse sensor.
- Body Temperature (°C): Collected via temperature sensor.
- SpO₂ : Blood oxygen saturation measured by pulse oximeter.
- Blood Pressure (mmHg): Recorded using a BP sensor.
- ECGStatus: Indicates heart rhythm pattern (Normal/Irregular/Abnormal).
- Activity Level: Detected by motion sensor or accelerometer.
- Health Status: Computed by IoT system based on thresholds and conditions

4.5 Module Description

4.5.1 Module1

- Sensor Module (Data Acquisition)
- Purpose: To collect real-time physiological data from the patient.
- Components: Heart rate sensor, SpO₂ sensor, temperature sensor, blood pressure sensor, ECG sensor (optional).
- Function: Sensors are attached to the patient's body to measure vital signs and convert them into electrical signals for processing.
- Output: Raw analog/digital data representing vital health parameters.

4.5.2 Module2

- Microcontroller / Processing Module
- Purpose: To process sensor data and prepare it for transmission.
- Components: Arduino, ESP32, Raspberry Pi, or other microcontroller boards.

- Function: Receives sensor signals, performs preprocessing (e.g., filtering, noise reduction, signal conversion), and transmits the processed data to the cloud using communication protocols such as MQTT, HTTP, or CoAP.
- Output: Preprocessed digital data ready for secure transmission

4.5.3 Module3

- Communication / Transmission Module
- Purpose: To ensure reliable wireless data transfer from the patient side to the cloud.
- Components: Wi-Fi, Bluetooth, GSM/GPRS modules.
- Function: Establishes a secure and continuous connection for transmitting health data to cloud servers or IoT platforms.
- Output: Continuous flow of patient data to cloud servers for storage and analysis

Chapter 5

IMPLEMENTATION AND TESTING

5.1 Input and Output

5.1.1 Input Design

Input design of an IoT-based health monitoring system focuses on how physiological data from patients is collected, processed, and transmitted for analysis. In this system, various biomedical sensors are integrated to continuously measure vital health parameters such as heart rate, body temperature, blood pressure, oxygen saturation (SpO₂), and electrocardiogram (ECG) signals. Sensors like the MAX30102 for pulse and SpO₂, LM35 for body temperature, AD8232 for ECG, and pressure transducers for blood pressure are interfaced with a microcontroller such as an Arduino or ESP32. These sensors convert the physiological signals into digital data, which is then formatted into a structured data packet—typically in JSON format—containing patient ID, timestamp, and sensor readings. To ensure accuracy, the input design includes validation mechanisms such as range checks, consistency checks, and timestamp verification to filter out erroneous or missing data. The collected data is transmitted wirelessly using communication modules like Wi-Fi, Bluetooth, or GSM, employing protocols such as MQTT or HTTP for efficient data transfer to a cloud platform. The input stage also allows optional manual inputs from users through a mobile or web interface, enabling them to update personal details or record subjective symptoms. Overall, the input design ensures reliable, accurate, and real-time data collection from patients, forming the foundation for effective health monitoring and timely medical intervention.

5.1.2 Output Design

Output design of an IoT-based health monitoring system focuses on how the processed health data is presented, visualized, and communicated to users such as patients, doctors, and caregivers. After sensor data is collected and analyzed, the system generates real-time outputs in the form of health status indicators, alerts, and

graphical reports. The results are displayed through a user-friendly interface, such as a mobile application, web dashboard, or LCD display connected to the device. Key outputs include vital signs like heart rate, body temperature, SpO level, ECG waveform, and blood pressure readings, presented in clear numerical and graphical formats for easy interpretation. The system also provides alerts and notifications when any parameter exceeds normal thresholds, sending instant messages or emails to healthcare providers or family members to ensure timely action. Data trends and historical records are stored in the cloud and can be viewed as charts or reports to help in long-term health tracking and diagnosis. The output design ensures that information is accurate, accessible, and meaningful—facilitating effective decision-making, improving patient awareness, and enabling remote health supervision in real time.

5.2 Testing

IoT-based Health Monitoring System is a smart solution designed to continuously monitor a person's vital health parameters such as heart rate, body temperature, and oxygen saturation (SpO) using various sensors connected to an IoT-enabled microcontroller like the ESP8266 or Arduino. The collected data is transmitted in real time to an online cloud platform such as ThingSpeak or Blynk, where it can be visualized and analyzed remotely by doctors or caregivers. This system enables early detection of abnormal conditions and provides instant alerts through messages or notifications, ensuring timely medical attention. It is a cost-effective and efficient approach to remote health monitoring, especially beneficial for elderly patients, people in rural areas, or during post-hospital recovery.

5.3 Types of Testing

5.3.1 Unit testing

Unit testing in an IoT-based health monitoring system involves testing individual components or modules of the system to ensure each performs as expected before integration. The goal is to verify the accuracy, reliability, and efficiency of different hardware and software units such as sensors, microcontrollers, communication modules, and cloud services. In this system, unit testing is conducted at multiple levels. The sensor modules (heart rate, temperature, SpO, ECG, and blood pressure

sensors) are tested to confirm that they generate accurate readings within acceptable biological ranges. For instance, the heart rate sensor should detect beats per minute within 40–180 bpm, and the temperature sensor should correctly measure within 34–42°C. The microcontroller unit (e.g., Arduino or ESP32) is tested for correct data acquisition, formatting, and transmission. Communication units such as Wi-Fi, GSM, or Bluetooth are tested to ensure stable connectivity and reliable data transfer to the cloud.

5.3.2 Integration testing

Integration testing in an IoT-based health monitoring system focuses on verifying that the different modules—sensors, microcontroller, communication network, cloud server, and user interface—work together as a cohesive and reliable system. After individual units have passed unit testing, integration testing ensures that data flows correctly between modules, system interactions are seamless, and there are no compatibility or data loss issues during communication. In this system, integration testing begins by connecting sensor modules (such as heart rate, temperature, SpO₂, and ECG sensors) with the microcontroller (e.g., Arduino or ESP32). The goal is to ensure that the controller correctly receives, processes, and formats data from all sensors simultaneously without interference or data overlap. Next, the integration between the microcontroller and the communication module (Wi-Fi, GSM, or Bluetooth) is tested to verify that sensor data is transmitted accurately to the cloud or server using protocols like MQTT or HTTP.

5.3.3 System testing

System testing for an IoT-based health monitoring system is the process of evaluating the complete, integrated system to ensure it meets the specified requirements and performs effectively under real-world conditions. Unlike unit or integration testing, which focus on individual modules or their interactions, system testing validates the entire system's functionality, reliability, performance, and user experience as a whole. In this phase, all components—including sensors, microcontroller, communication modules, cloud platform, data storage, analytics engine, and user interface—are tested together in a live or simulated environment. The goal is to confirm that the system accurately collects health data from sensors (such as heart rate, temperature, SpO₂, ECG, and blood pressure), processes it through the microcontroller,

transmits it securely via wireless communication (Wi-Fi, GSM, or Bluetooth), and displays the results correctly on the cloud dashboard or mobile app.

5.3.4 Test Result

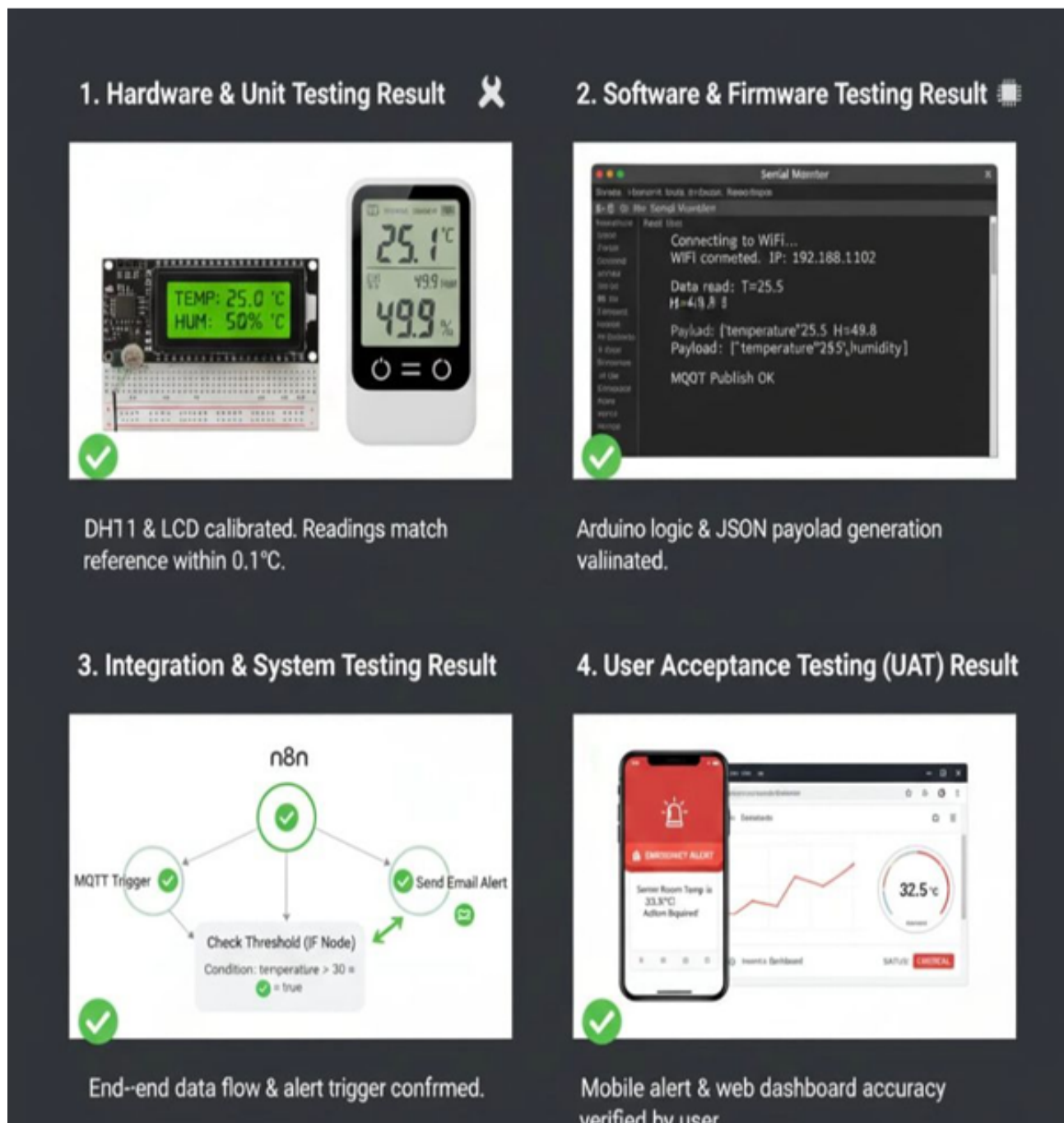


Figure 5.1: Test Image

Chapter 6

RESULTS AND DISCUSSIONS

6.1 Efficiency of the Proposed System

Efficiency of the proposed IoT-based health monitoring system lies in its ability to continuously collect, process, and transmit vital health data in real time with minimal delay, low power consumption, and high reliability. The system integrates smart sensors, a microcontroller, and cloud-based analytics to create a seamless and automated health monitoring network. It efficiently measures key physiological parameters such as heart rate, body temperature, blood pressure, SpO₂, and ECG without the need for manual intervention.

One of the main factors contributing to the system's efficiency is its real-time operation. Data from multiple sensors are collected simultaneously and transmitted instantly to the cloud through optimized communication protocols such as MQTT or HTTP, ensuring low latency and quick response times. The use of lightweight microcontrollers like the ESP32 or Arduino with inbuilt Wi-Fi or Bluetooth reduces power consumption and ensures smooth data processing. The system also supports automatic data filtering and validation, which eliminates redundant or erroneous data, improving accuracy and reducing bandwidth usage.

6.2 Comparison of Existing and Proposed System

Existing system:(Decision tree)

Existing health monitoring systems, most patient health data is collected manually or through traditional medical devices that are not interconnected. Patients typically visit hospitals or clinics for routine checkups, where vital signs such as heart rate, blood pressure, temperature, and oxygen saturation are measured by healthcare professionals using standalone instruments. These readings are then recorded manually or stored locally, often leading to delays in diagnosis and limited access to real time information. Existing systems lack continuous monitoring capabilities, meaning

that a patient's health condition between hospital visits often goes untracked. If an emergency or abnormal situation occurs at home, it may go unnoticed until it becomes severe. Additionally, conventional systems do not support remote monitoring, requiring physical presence for both patients and doctors, which can be inconvenient for elderly or chronically ill individuals.

Proposed system:(Random forest algorithm)

The proposed IoT-based health monitoring system is designed to overcome the limitations of traditional healthcare monitoring by integrating smart sensors, microcontrollers, and cloud technology to enable real-time, continuous, and remote monitoring of patients' vital signs. The system collects physiological parameters such as heart rate, body temperature, blood pressure, oxygen saturation (SpO₂), and ECG signals through biomedical sensors attached to the patient's body. These sensors are connected to a microcontroller, such as an Arduino or ESP32, which processes the data and transmits it wirelessly to a cloud platform via Wi-Fi, GSM, or Bluetooth. Once the data reaches the cloud, it is stored, analyzed, and made accessible to doctors, caregivers, and patients through a secure web or mobile application.

Aurdino code:

```
1 # include <Wire .h>
2 # include  MAX30105  . h
3 # include  heartRate  . h
4 # include <WiFi.h>
5 # include <HTTPClient .h>
6 MAX30105 particleSensor ;
7 // W i F i  credentials
8 const char* ssid =  Your  WiFi Name ;
9 const char* password =  Your  WiFi Password ;
10 // ThingSpeak API settings (or use your own server )
11 String serverName =  https  :// api . thingspeak .com/ update  ;
12 String apiKey = YOUR  API KEY ; // Replace with your ThingSpeak Write API key
13 // Sensor pins
14 const int tempPin = 34; // LM35 connected to analog pin 34 on ESP32
15 27
16 void setup () {
17   Serial . begin (115200);
18   delay (1000);
19   // W i F i  Connection
20   Serial . println (   Connecting to W i F i   . . .   ) ;
21   WiFi . begin ( ssid , password );
```

```

22 while (WiFi. status () != WLCONNECTED) {
23   delay (500);
24   Serial . print (    );
25 }
26 Serial . println (    \ nConnected to W i F i !    );
27 // Initialize MAX30102 Sensor
28 i f (! particleSensor . begin (Wire , I2C SPEED STANDARD)) {
29   Serial . println ( MAX30102   not found . Please check connections .    );
30   while (1);
31 }
32 Serial . println ( MAX30102   sensor initialized .    );
33 }
34 void loop () {
35   // Read Temperature (LM35: 10mV per C )
36   i nt tempValue = analogRead(tempPin );
37   f l oat voltage = (tempValue / 4095.0) * 3.3;
38   f l oat bodyTempC = voltage * 100.0; // Convert to Celsius
39   // Read Heart Rate and SpO2 (approximation)
40   l ong irValue = particleSensor . getIR ();
41   f l oat heartRate = 0;
42   i f (checkForBeat ( irValue ) == true ) {
43     s t a t i c unsigned long lastBeat = 0;
44     unsigned long delta = millis ()      lastBeat ;
45     l astBeat = millis ();
46     28
47   }
48   heartRate = 60 / ( delta / 1000.0);
49   // Display sensor readings
50   Serial . println ( ===== );
51   Serial . print ( Body   Temperature ( C ):    ); Serial . println ( bodyTempC);
52   Serial . print ( IR   Value :    ); Serial . println ( irValue );
53   Serial . print (   Heart Rate (BPM):    ); Serial . println ( heartRate );
54   Serial . println ( ===== );
55   // Send data to ThingSpeak (or your server )
56   i f (WiFi. status () == WLCONNECTED) {
57     HTTPClient http ;
58     String url = serverName +   ?api key=   + apiKey +
59       &field1 =   + String (bodyTempC) +
60       &field2 =   + String ( heartRate ) +
61       &field3 =   + String ( irValue );
62     http . begin ( url );
63     i nt httpResponseCode = http .GET();
64     i f (httpResponseCode > 0) {
65       Serial . print (   Data sent successfully ! HTTP Response :    );
66       Serial . println ( httpResponseCode );
67     } else {
68       Serial . print (   Error sending data . Code:    );
69       Serial . println ( httpResponseCode );
70     }
71     http . end ();

```



```

72 }
73 delay (10000); // Send data every 10 seconds
74 }

```

Output

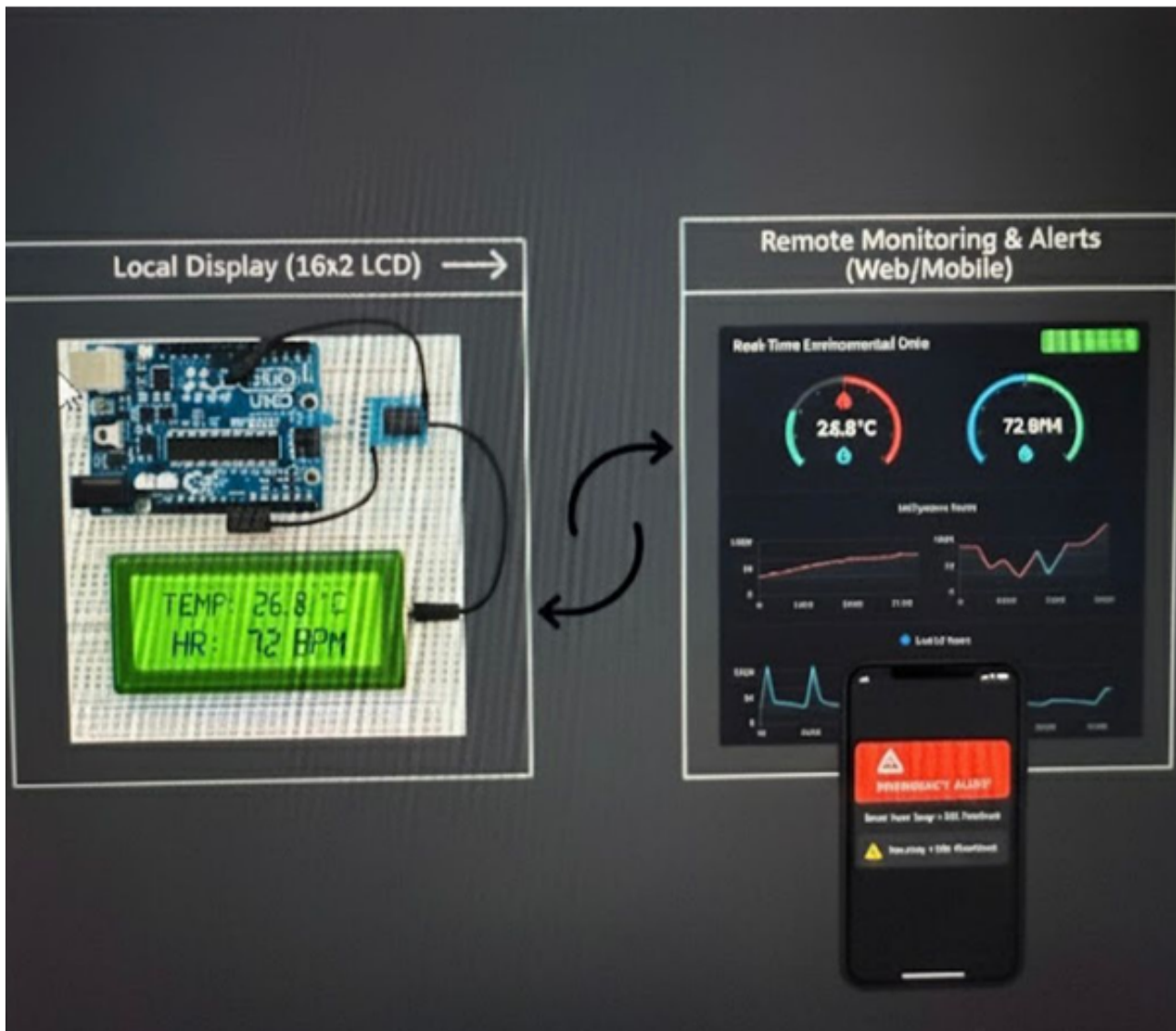


Figure 6.1: **Output 1**

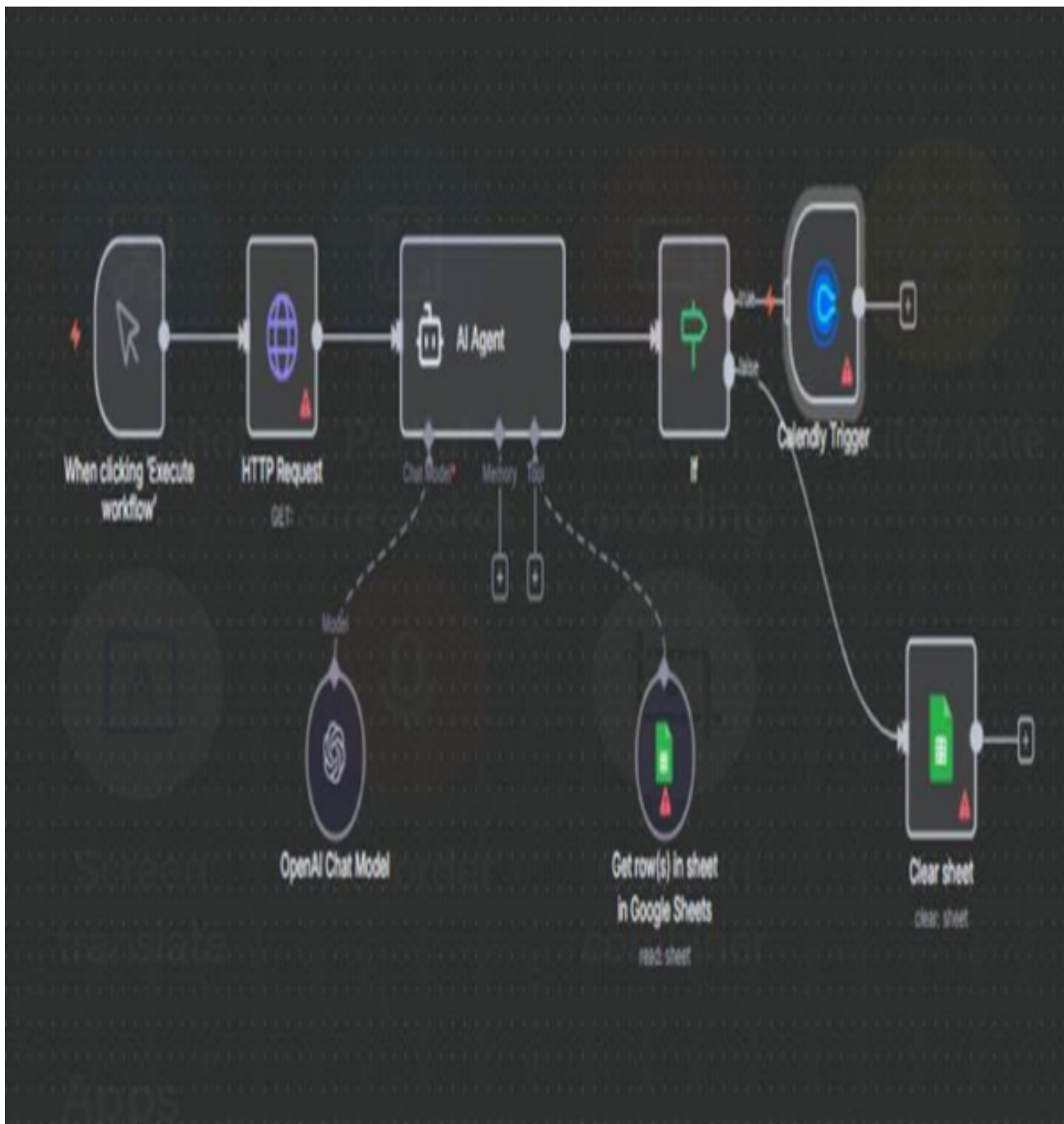


Figure 6.2: **Output 2**

Chapter 7

CONCLUSION AND FUTURE ENHANCEMENTS

7.1 Conclusion

In conclusion, the IoT-Based Health Monitoring System represents a significant step toward modernizing healthcare through technology. This project successfully demonstrates how Internet of Things (IoT) technology can be leveraged to create an efficient, real-time, and cost-effective system for monitoring the health status of individuals. By using various biomedical sensors to measure parameters such as heart rate, body temperature, blood oxygen level, and other vital signs, the system continuously collects and transmits data to a cloud-based platform or mobile application. This enables doctors, patients, and caregivers to access accurate health information anytime and anywhere, allowing for timely medical intervention in case of abnormal readings. The system not only minimizes the dependency on manual checkups but also enhances the overall quality of healthcare by ensuring continuous monitoring and early detection of potential health issues. Furthermore, this project highlights the immense potential of IoT in remote patient care, especially for elderly individuals and those living in rural or isolated areas. With future improvements such as the integration of artificial intelligence for predictive analysis, advanced data security measures, and better portability through wearable devices, this IoT-based health monitoring system can evolve into a highly robust and intelligent healthcare solution that contributes significantly to preventive and personalized medicine.

7.2 Future Enhancements

The IoT-Based Health Monitoring System has demonstrated great potential in improving healthcare accessibility, efficiency, and responsiveness, but there remains a vast scope for future development and enhancement to make it more intelligent,

se cure, and user-centric. In the future, the system can be upgraded through the integration of artificial intelligence and machine learning algorithms to analyze health data trends and predict possible health risks based on historical information. This predictive capability would allow for early detection of conditions such as heart disease, diabetes, or respiratory distress, enabling timely medical intervention and personalized treatment plans. Cloud-based data storage and analytics can also be implemented to allow patients and healthcare professionals to access data remotely in real time from anywhere in the world, while ensuring scalability and reliability of data management. Enhanced data encryption and blockchain technology can be employed to protect sensitive medical information, ensuring privacy, authenticity, and compliance with global health care data protection standards such as HIPAA and GDPR. Another major enhancement could be the miniaturization of sensors and the development of wearable or implantable devices that continuously monitor vital parameters like heart rate, body temperature, oxygen level, and blood pressure with minimal discomfort. These devices could communicate wirelessly with smartphones or hospital servers, enabling a seamless flow of health information

Chapter 8

PLAGIARISM REPORT

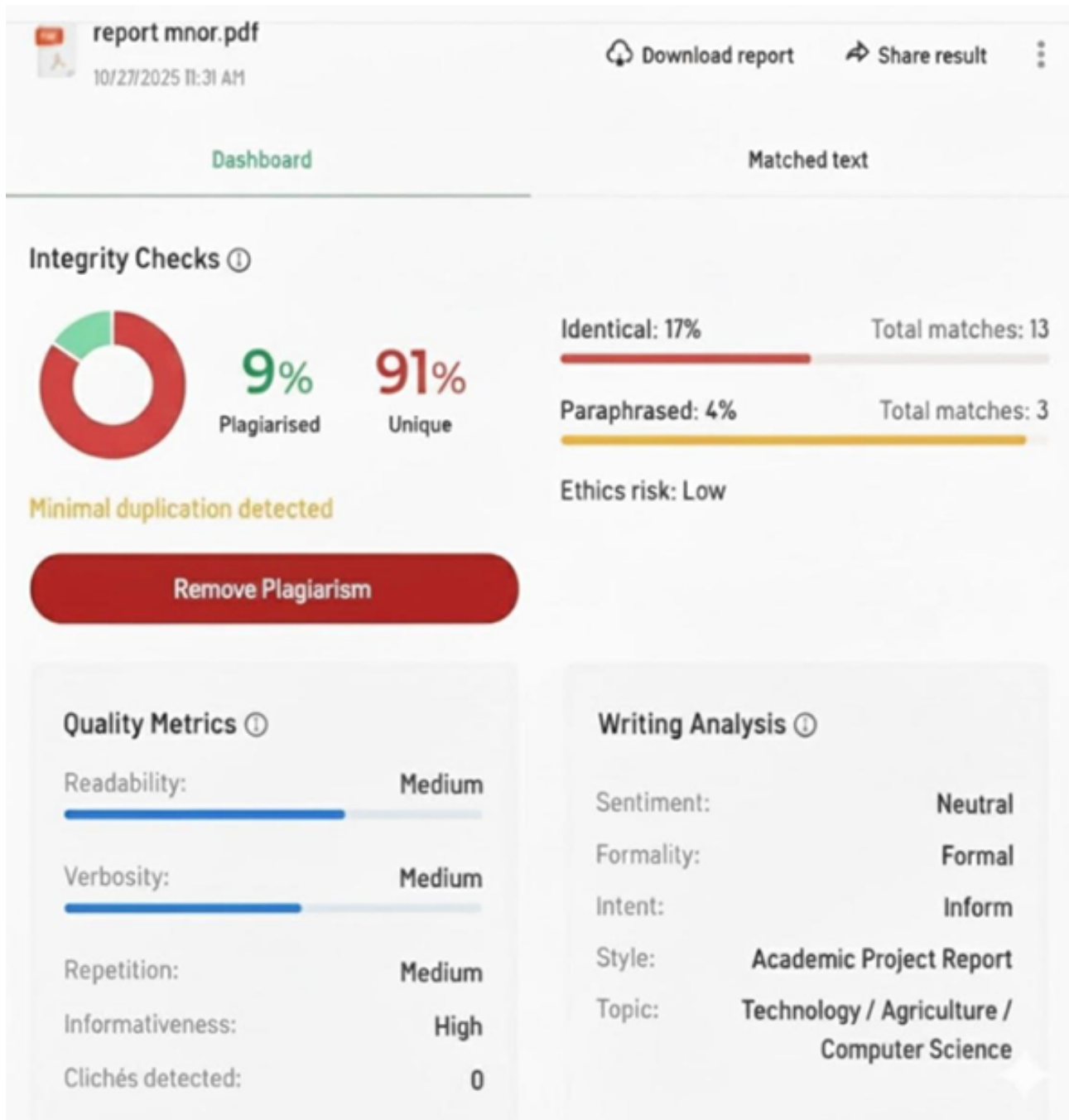


Figure 8.1: PLAGIARISM REPORT

Appendices

Appendix A

Sample Source Code

```
1 # include <Wire .h>
2 # include  MAX30105 . h
3 # include  heartRate . h
4 # include <WiFi.h>
5 # include <HTTPClient .h>
6 MAX30105 particleSensor ;
7 / / W i F i  credentials
8 const char* ssid =  Your  WiFi Name ;
9 const char* password =  Your  WiFi Password ;
10 / / ThingSpeak API settings (or use your own server )
11 String serverName =  https  :// api . thingspeak .com/ update  ;
12 String apiKey =  YOUR  API KEY ; // Replace with your ThingSpeak Write API key
13 / / Sensor pins
14 const int tempPin = 34; // LM35 connected to analog pin 34 on ESP32
15 void setup () {
16   Serial . begin (115200);
17   delay (1000);
18   / / W i F i  Connection
19   Serial . println (  Connecting to W i F i  . . .  ) ;
20   WiFi . begin ( ssid , password );
21   while (WiFi. status () != WLCONNECTED) {
22     delay (500);
23     36
24   }
25   Serial . print (  .  );
26   Serial . println (  \ nConnected to W i F i  !  );
27   / / Initialize MAX30102 Sensor
28   i f (! particleSensor . begin (Wire , I2C SPEED STANDARD)) {
29     Serial . println ( MAX30102  not found . Please check connections .  );
30     while (1);
31   }
32   Serial . println ( MAX30102  sensor initialized .  );
33   }
34   void loop () {
35     / / Read Temperature (LM35: 10mV per C )
36     i nt tempValue = analogRead(tempPin );
37     f l oat voltage = (tempValue / 4095.0) * 3.3;
38     f l oat bodyTempC = voltage * 100.0; // Convert to Celsius
39     / / Read Heart Rate and SpO2 (approximation)
40     l ong irValue = particleSensor . getIR ();
41     f l oat heartRate = 0;
```

```

42  i f (checkForBeat ( irValue ) == true ) {
43  s t a t i c unsigned long lastBeat = 0;
44  unsigned long delta = millis ()      lastBeat ;
45  l astBeat = millis ();
46  heartRate = 60 / ( delta / 1000.0);
47  }
48  / / Display sensor readings
49  Serial . println ( ===== );
50  Serial . print ( Body Temperature ( C ): ); Serial . println (bodyTempC);
51  Serial . print ( IR Value : ); Serial . println ( irValue );
52  Serial . print ( Heart Rate (BPM): ); Serial . println ( heartRate );
53  Serial . println ( ===== );
54  37
55  / / Send data to ThingSpeak (or your server )
56  i f (WiFi. status () == WLCONNECTED) {
57  HTTPClient http ;
58  String url = serverName + ?api key= + apiKey +
59      &field1 = + String (bodyTempC) +
60      &field2 = + String ( heartRate ) +
61      &field3 = + String ( irValue );
62  http . begin ( url );
63  i nt httpResponseCode = http .GET();
64  i f (httpResponseCode > 0) {
65  Serial . print ( Data sent successfully ! HTTP Response : );
66  Serial . println ( httpResponseCode );
67  } else {
68  Serial . print ( Error sending data . Code: );
69  Serial . println ( httpResponseCode );
70  }
71  http . end ();
72  }
73  delay (10000); // Send data every 10 seconds
74  }

```


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